

Intro to Dart

DMIT 2504 – WEEK 1 DAY 2



Announcements

Expectations - What I expect from you

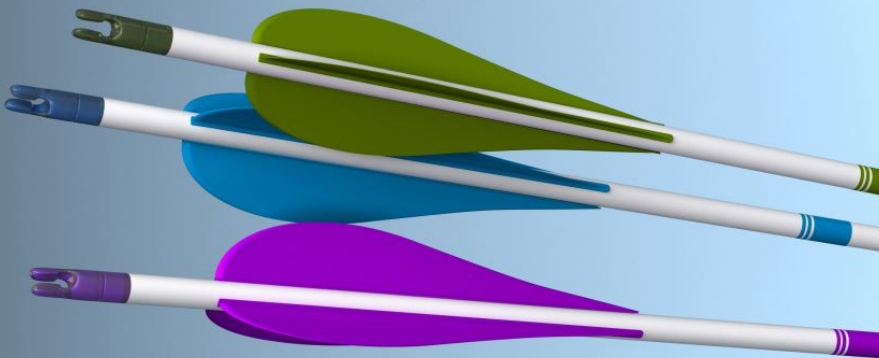
1. No Late Assignments
2. No Cheating
3. Be a good classmate
4. Don't go to stupid sites in class
5. Show up to class

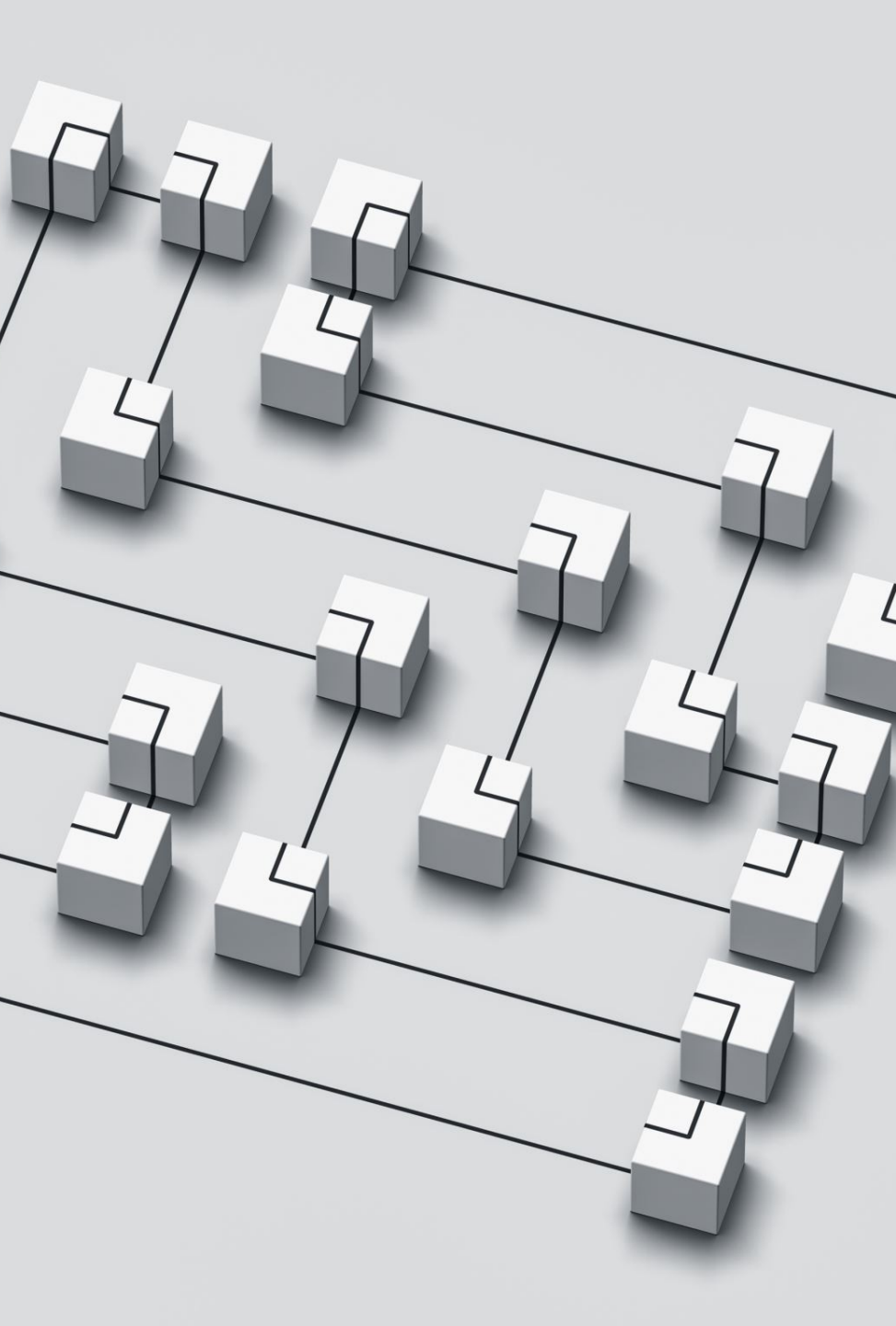
OBJECTIVES

After participating in this session, students should be able to ...

- Describe the Dart programming language
- Create a Dart console application project
- Run a Dart console application
- Capture input and display output in a Dart console application

The accompanying Dart files can be found and downloaded from the course LMS site

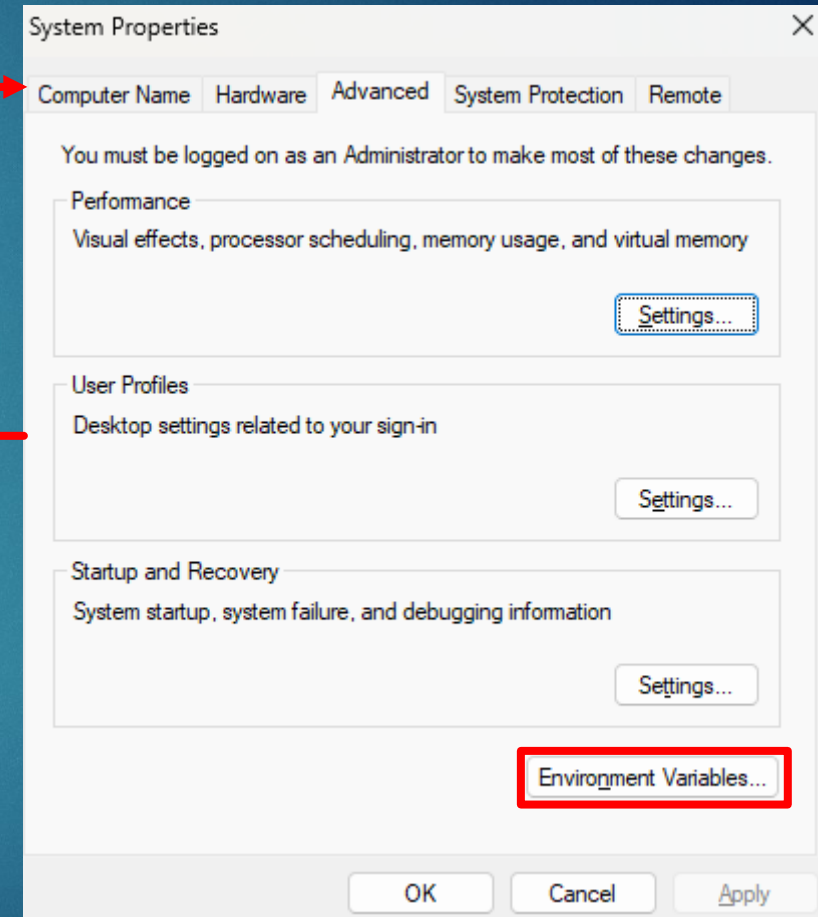
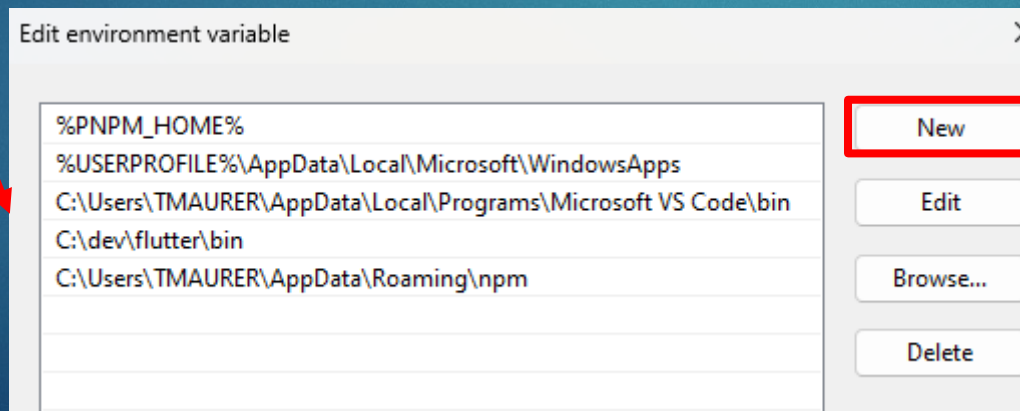
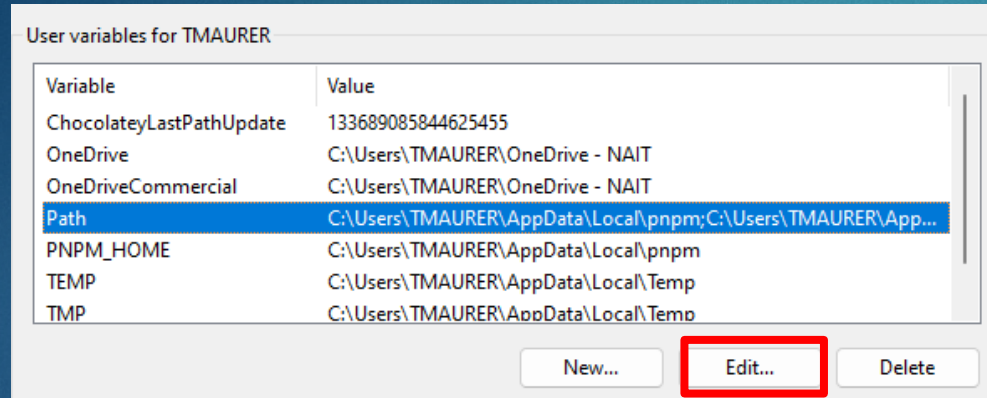
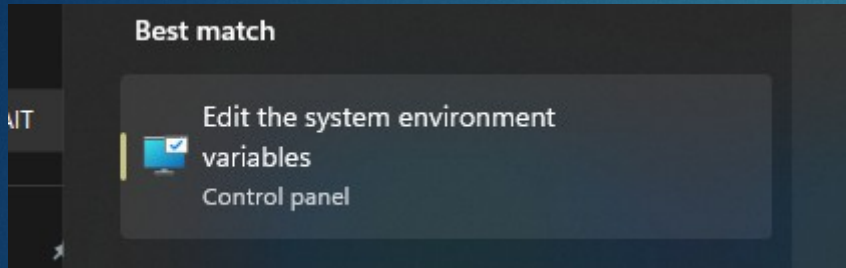




Question from last class: What is the PATH?

PATH is an environment variable on several operating systems Microsoft Windows, specifying a set of directories where executable programs are located.

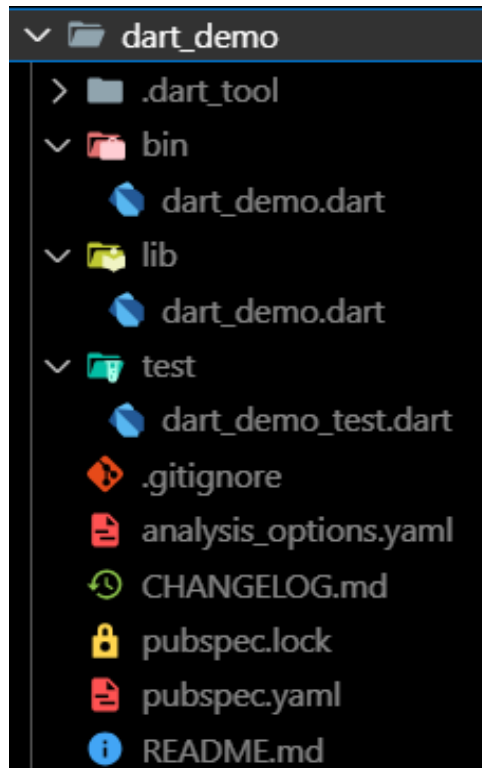
How to PATH?



DART LANGUAGE

- ▶ Dart, created by Google, is a client-optimized programming language for fast apps on any platform
 - ▶ [Dart Programming Language](#)
- ▶ Designed for building mobile, desktop, server, and web applications
- ▶ Ahead-of-time (AOT) compiled to native code for fast execution
- ▶ Learn quickly with DartPad, an online tool for running Dart code directly in the browser
 - ▶ [DartPad](#)
- ▶ Code style and conventions are documented online
 - ▶ [Effective Dart | Dart](#)

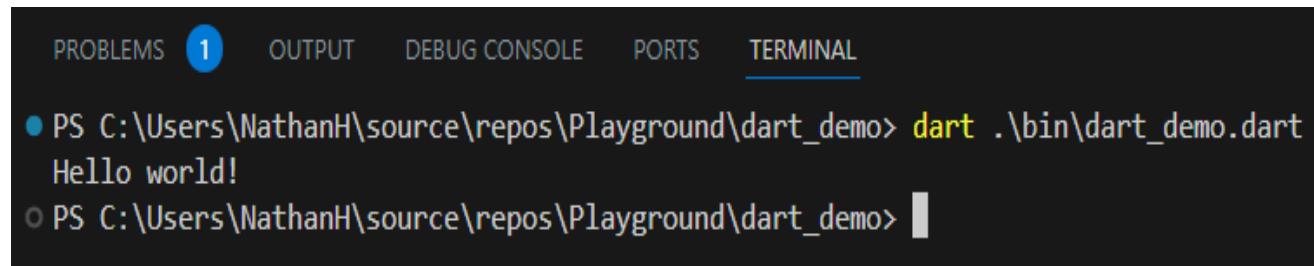
Dart Console Application



- ▶ Create a Dart Console Application in VS Code
 - ▶ Ctrl+Shift+P `Dart: NewProject` `Console Application`
 - ▶ Follow the appropriate naming convention
- ▶ Dart project directory structure:
 - ▶ bin -the main executable is contained here (main method)
 - ▶ lib - additional library code for your project can be placed here
 - ▶ test -unit tests can be placed here

RUNNING THE APPLICATION

- ▶ Open a terminal
- ▶ Navigate to the project directory (use the `cd` command)
- ▶ Execute your Dart program using the `dart` command
 - ▶ `dart.\bin\dart_demo.dart`



A screenshot of a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'PORTS', and 'TERMINAL'. The 'TERMINAL' tab is active and underlined. Below the tabs, the terminal shows a PowerShell prompt 'PS C:\Users\NathanH\source\repos\Playground\dart_demo>' followed by the command 'dart .\bin\dart_demo.dart'. The output 'Hello world!' is displayed on the next line. A second prompt line is visible below, showing the same directory path and a cursor.

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE PORTS TERMINAL
PS C:\Users\NathanH\source\repos\Playground\dart_demo> dart .\bin\dart_demo.dart
Hello world!
PS C:\Users\NathanH\source\repos\Playground\dart_demo> |
```

DART CONSOLE OUTPUT

- ▶ You can use the `stdout` object from the `dart:io` library to display text on the console
 - ▶ `write()` will write text and leave the cursor on the same line
 - ▶ `writeln()` will write text and move the cursor to the next line
- ▶ While Dart allows both single and double quotes to be used to delimit strings, single quotes are preferred
- ▶ Interpolate variables using a `$` or `${}` for expressions or `.accessor`

```
dart_demo > bin > dart_demo.dart > ...  
1  import 'dart:io';  
2  
   Run | Debug  
3  void main(List<String> arguments) {  
4      stdout.writeln('Hello world!');  
5  }
```

DART CONSOLE INPUT

- ▶ You can use the `stdin` object from the `dart:io` library to read in a string from the console
 - ▶ `stdin.readLineSync()` will read in the next line as a string
- ▶ Types can be used to parse strings into other types
 - ▶ `int.parse()`
 - ▶ `double.parse()`

```
dart_demo > bin > dart_demo.dart > ...  
1  import 'dart:io';  
2  
   Run | Debug  
3  void main(List<String> arguments) {  
4      stdout.write('Enter your name: ');  
5      final name = stdin.readLineSync();  
6      stdout.writeln('Hello $name, how are you?');  
7  }
```

ADD LIBRARY FUNCTIONS

- ▶ You can create functions in a library file and import those functions into your program

```
dart_demo > lib > dart_demo.dart > ...
1  import 'dart:io';
2
3  /// Returns a [String]
4  ///
5  /// Displays the passed in [message] as the prompt for the user.
6  String prompt(String message) {
7    stdout.write(message);
8    String? input = stdin.readLineSync();
9    return input ?? '';
10 }
```

```
dart_demo > bin > dart_demo.dart > ...
1  import 'dart:io';
2
3  import 'package:dart_demo/dart_demo.dart';
4
5  Run | Debug
6  void main(List<String> arguments) {
7    final name = prompt('Enter your name: ');
8    stdout.writeln('Hello $name, how are you?');
9  }
```

SUMMARY

- ▶ Dart is a client-optimized programming language for targeting any platform
- ▶ VS Code can be used to quickly create Dart projects
- ▶ The `dart:io` library has classes for displaying output and capturing input from the console

```
/// Returns an [int]
///
/// Will continue to prompt the user until a valid
/// Displays the passed in [message] as the prompt
int promptInt(String message) {
    bool isValid = false;

    while (!isValid) {
        String input = prompt(message);
        try {
            return int.parse(input);
        } catch (e) {
            stdout.writeln('Invalid input. Please enter a
        }
    }
}
```